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The history of chicken and other bird exploitation in Thailand: Preliminary analysis of bird remains from four archaeological sites

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Abstract

The origin of the domestication of chicken *Gallus gallus domesticus* is still a subject of debate. It principally originates from the red junglefowl *G. gallus*, which is distributed throughout Southeast Asia and South China. However, the prehistoric exploitation of chicken and red junglefowl in Southeast Asia has remained unclear due to a small number of samples for bone analysis. In this study, we analysed bird remains from four archaeological sites in Thailand: Ban Hua Ud (4,000–3,000 BP), Long Long Rak (1,900–1,600 BP), Ban Non Wat (3,750–1,500 BP), and Keed Kin (fifth–11th Century A.D.). Bones from pheasants/fowls (Phasianidae), storks (Ciconiidae), cormorants (Phalacrocoracidae), crows (Corvidae), and passerines (Passeriformes) were found. A small number of stork bones were found at three lowland sites, with none found at Long Long Rak. Cormorant bones were dominant at Ban Hua Ud but were only found at that site. Pheasant/fowl bones were found at all sites except Ban Hua Ud and dominated the number of identified specimens, suggesting that they were the most commonly used bird, at least after the Iron Age. Morphological bone identification criteria for the 24 Phasianidae species in Thailand have not been established; ancient DNA analysis is required to differentiate chicken/red junglefowl bone from other indigenous pheasants/fowls. It is obvious that mitochondrial DNA remained unchanged in the early stages of the domestication process. Further studies combining ancient DNA and morphological analyses are required to judge whether all of the candidate chicken bones are indeed from red junglefowl or chicken. Although these results are regionally and temporally restricted, they give important quantitative information on bird exploitation during the Neolithic and early historic Dvaravadi periods in Thailand.

KEYWORDS

bird exploitation, domestic chicken, morphological identification, Thailand, wild bird

1 | INTRODUCTION

Chickens (*Gallus gallus domesticus*) are one of the most widespread domestic animals worldwide; they are primarily used as a source of food, providing meat and eggs to every continent and even the remote islands. In the early stages of chicken domestication, they appear to have been used for religious activities, cockcrowing, and cockfighting (Serjeantson, 2009). The domestic chicken principally originates from the red junglefowl *G. gallus*, which is distributed throughout Southeast Asia and South China. Genetic studies of chickens have proposed multiple origins of domestication, including Southeast Asia, East Asia, and South Asia (Fumihito et al., 1994; Liu et al., 2006; Miao et al., 2013).

The origin of chicken domestication is still a subject of debate. Ancient DNA analysis of Phasianidae bones from Nanzhuangtou

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(Hebei Province, China) indicated chicken mtDNA haplotypes. This suggests that the wild distribution area of red junglefowl was broader in the early Holocene and that domestic chicken farming began in approximately 10,000 BP in North China (Xiang et al., 2014). However, other studies have challenged the discovery of early Holocene chicken bones in North China (Eda, Lu et al., 2016; Peters, Lebrasseur, Deng, & Larson, 2016). Eda, Lu et al. (2016) revealed that candidate chicken bones were rare in Neolithic and early Bronze Age sites in central and North China, whereas Peters et al. (2016) discussed that the post-Pleistocene climatic conditions in North China were unsuitable for the wild distribution of red junglefowl. Both studies concluded that domestic chickens and red junglefowl were neither widely kept nor distributed in North China during the early and middle Holocene. It would be reasonable to assume that chicken domestication began in regions where wild red junglefowl were distributed at that time.

Prehistoric exploitation of chicken and red junglefowl in Southeast Asia (where wild junglefowl exists today) has remained unclear, because bird bone analyses are scarce (West, 1991; Higham & Kijngam, 1984, 2012; Conrad, 2015). Previous DNA studies (Fumihito et al., 1996; Storey et al., 2012) have suggested that Thailand might be one of the origins of domestication. However, whereas two bones from junglefowl/domestic fowl (*G. gallus*) were found at Khok Phanom Di (Neolithic; West, 1991), and complete chicken skeletons were reported at Ban Non Wat (Bronze Age; Higham & Kijngam, 2012) and Ban Na Di (Bronze Age; Higham & Kijngam, 1984), few zooarchaeological bird remains have been analysed from this region.

This paper aims to (a) provide the results of a preliminary analysis of 117 bird remains from four archaeological sites in Thailand (dating from the Neolithic and Historic periods) and (b) discuss the analysis from a historical perspective with reference to other zooarchaeological bird finds in Thailand, to better understand chicken exploitation in the region.

2 | MATERIALS AND METHODS

2.1 | Study sites

Samples (117) were taken from four archaeological sites of different time periods and analysed. The four sites are Ban Hua Ud, Long Long Rak, Ban Non Wat, and Keed Kin (Figure 1).

2.1.1 | Ban Hua Ud (Neolithic, 4000–3000 BP)

Ban Hua Ud is located in Snam Kree Subdistrict, Muang District, Suphan Buri Province, Central Thailand. The site was occupied by multiple cultural groups from prehistoric to historic periods, ranging from 4,000 to 400 years ago (Chantakul, 2016; Department of Archaeology, 2009). It functioned as both a habitation and burial site. A previous study of the site (Chantakul, 2016; Department of Archaeology, 2009) revealed 37 identified species, including cattle (*Bos* sp.), buffalo (*Bubalus bubalis*), deer (*Cervus* sp.), pig (*Sus scrofa*), dog (*Canis* sp.), barking deer (*Muntiacus muntjak*), monitor lizard (*Varanus* sp.), elephant (*Elephas maimus*), rodents (Rodentia), many species of fish (e.g., *Channa* sp., *Clarias* sp., *Pangasianodon hypophthalmus*, *Wallago attu*), turtle (Testudines), snapping turtle (Trionychidae), crocodile (*Crocodylus sianmesis*), and bird. From this site, nine bird remains were studied.

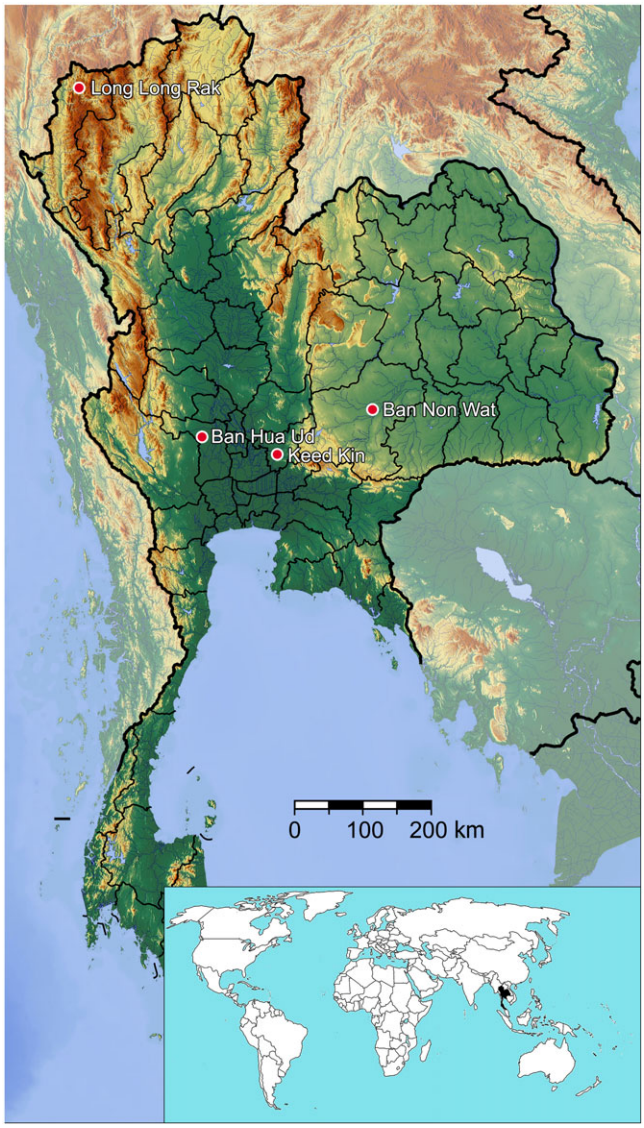


FIGURE 1 Locations of the studied sites [Colour figure can be viewed at wileyonlinelibrary.com]

2.1.2 | Long Long Rak Coffin Cave (Iron Age, 1900–1600 BP)

The Long Long Rak Cave is located in Moo 1, Ban Tham Lod, Tham Lod Subdistrict, highland Pang Mapha District, Mae Hong Son Province, northwest Thailand. The site is surrounded by mixed deciduous and bamboo forests. The dry cave is located on top of limestone cliffs about 735 m above average sea level and is about 300 m from the course of the Lang River.

The mouth of the cave opens up to a complex room approximately 15 m into the cave, which is divided into three large chambers. The chambers are referred to as Chambers A, B, and C during excavation and were used as family burial places (Shoocongdej, Chintakanon, Pureepatpong, Phunmijumnong, & Saminpanya, 2016). The bone samples are from Chamber A1, which was used as a main cemetery; 19 log coffins were found inside. Faunal remains, such as pig (*S. scrofa*), dog

(*Canis* sp.), and bird (reported as chicken), were found in the coffins. Carbon-14 dating results from the 18 samples, collected from coffin heads and lacquers and dated between 1960 ± 30 (Beta-398272) and 1,636 ± 44 BP (UBA-27247), revealed that it had continuously been used for more than 300 years. From this site, 21 bird remains were studied.

2.1.3 | Ban Non Wat (Neolithic–Iron Age, 3,750–1,500 BP)

Ban Non Wat is a moated prehistoric site located in the Phon Songkram Subdistrict, Noen Sung District, Nakon Ratchasima Province, northeast Thailand. In comparison with other sites, the excavation of this site took place over several seasons. Faunal remains included domesticated animals such as cattle (*Bos* sp.), buffalo (*Bubalus bubalis*), pig (*Sus scrofa*), and dog (*Canis* sp.), as well as wild animals, such as deer (*Cervus* sp.) and wild pig. In addition, a complete chicken was found in a Bronze Age burial (ca. 800 BC) and identified as a domesticated chicken (Higham & Kijngam, 2012). The site functioned as a habitation and burial site over multiple periods. The site covers three periods: the Neolithic (3750–3100 BP), Bronze Age (3100–2500 BP), and Iron Age (2500–1500 BP; Higham, 2011; Higham & Higham, 2009).

From this site, 44 bird remains were studied. Five bones were from the Neolithic, seven were from the Late Neolithic–Early Bronze Age, three were from the Iron Age, and 29 were from the Neolithic–Iron Age. Six of the bones were previously analysed (Storey et al., 2012), and chicken mitochondrial DNA sequences were obtained from two of them.

2.1.4 | Keed Kin (Early Historic Dvaravadi, fifth–11th Century AD)

Keed Kin is located in Ban Koo Muang, Ban Mor Subdistrict, Ban Mor District, Saraburi Province, central Thailand. It is a rectangular site with double moats, which is typical of the Dvaravadi early historical period. The site functioned as a habitation; it was occupied through different periods between 1,200 and 2,500 years ago based on eight carbon-14 samples (Kachachiva et al., 2007). Faunal remains include cattle/buffalo (Bovidae), pig (*Sus* sp.), deer (*Cervus unicolor*), barking deer (*M. muntjak*), dog (*Canis* sp.), monitor lizard (*Varanus* sp.), turtle (Chelonian), and fish.

From this site, 43 bird remains were studied from the Dvaravadi period (1,200–1,400 years ago).

2.2 | Identification of bird remains

Naked-eye bird bone identification was conducted by one of the authors (M.E.), referring to modern osteological specimens at the Zooarchaeological Laboratory, Silpakorn University, Thailand. In order to confirm the identification results, we took photographs of some of the samples and compared them with osteological specimens stored at the Thailand Natural History Museum (TNHM) and Hokkaido University Museum, Japan. For each specimen, the developmental stage (mature or immature) and presence or absence of cut and/or manufacturing marks were recorded. The greatest length and breadth of the epiphyses were measured, referring to von den Driesch (1976). The developmental stage of each sample was defined as mature with smooth bone surface or immature with rough bone surface. Concerning broken bones, we also checked and recorded the presence or absence of medullary bone, which clearly shows that the bone was from a breeding female bird (Simkiss, 1961).

There are 24 Phasianidae (Tetraoninae and Phasianinae) species, including red junglefowl, in Thailand (Robson, 2013). As far as we know, there are no established morphological bone identification criteria for Phasianidae; therefore, the morphological identification of chicken/red junglefowl bones from the other 23 indigenous pheasants/fowls is a major challenge. On the other hand, it is possible to discriminate candidate chicken bones from nonchicken bones when they are femora or tarsometatarsi, according to the identification criteria of Eda and Inoué (2011); their study focused on the distinction of chicken/red junglefowl from Japanese indigenous pheasants (common pheasant *Phasianus colchicus* and copper pheasant *Syrmaticus soemmerringii*) and revealed that chickens/red junglefowl do not have pneumatic foramina on the greater trochanter (femur [Eda & Inoué, 2011]), nor a medial plantar crest (tarsometatarsus [Nishida & Hayashi, 1981]).

3 | RESULTS

3.1 | Ban Hua Ud

Nine bird remains were identified to the family level as cormorants (Phalacrocoracidae) and stork (Ciconiidae; Table 1, Figures 2); the number of identified specimens (NISP) was eight and one, respectively. The minimum number of individuals (MNI) was one for each taxon. All of the bones were mature, but a broken and healed profile was observed on a cormorant's coracoid. The size of one stork ulna studied (Bp [breadth of proximal end]: 20.4 mm, Data S1) was much larger than that of a painted stork (*Mycteria leucocephala*) specimen (Bp: 17.9 mm) stored at the TNHM and similar to that of black storks (*Ciconia nigra*, range: 19.3–20.9 mm, mean: 20.5 mm, $N = 5$). The cormorant bones were as large as a great cormorant specimen (*Phalacrocorax carbo*), owned by M.E. (EP-54). Neither cut nor manufacturing marks were observed on any of the bones.

3.2 | Long Long Rak Coffin Cave

Five of 21 bird remains were identified as pheasant/fowl (Phasianidae). The MNI for pheasant/fowl was two. The bones from pheasant/fowl consisted of hindlimb bones: four tarsometatarsi and one femur. There were no medial plantar crests (Nishida & Hayashi, 1981) on three tarsometatarsi, whereas it was unknown whether a crest was present or not at the distal fragment of one tarsometatarsus. Concerning the spur, it was obviously absent from one of the tarsometatarsi, and its presence was unknown in the rest of the samples. There were no pneumatic foramina at the greater trochanter (Eda & Inoué, 2011) of the femur. The sizes of the pheasant/fowl bones were similar to those of a modern female red junglefowl specimen, owned by one of the authors (P.A.). The rest of the bones consisted

TABLE 1 Taxa found from archaeological sites

	Archaeological site						
	Ban Hua Ud	Long Long Rak	Ban Non Wat				Keed Kin
Taxa	4,000–3,000 BP	1,900–1,600 BP	3,750–1,000 BP(in total*)	Neolithic	Late Neolithic–Early Bronze Age	Iron Age	1,500–900 BP
Order Galliformes							
Pheasant/fowl (Phasianidae)		5 (2)	16 (5)	4(1)	6(2)	3(2)	25 (6)
Order Pelecaniformes							
Cormorant (Phalacrocoracidae)	8 (1)						
Order Ciconiiformes							
Stork (Ciconiidae)	1 (1)		1 (1)		1 (1)		3 (1)
Order Passeriformes							
Crow (Corvidae)							2 (1)
Passerines (Passeriformes)			1 (1)				
Undetermined		16	26	1			13

*Including samples from Neolithic (5), Late Neolithic-Early Bronze Age (7), Iron Age (3), and Neolithic-Iron Age (29).

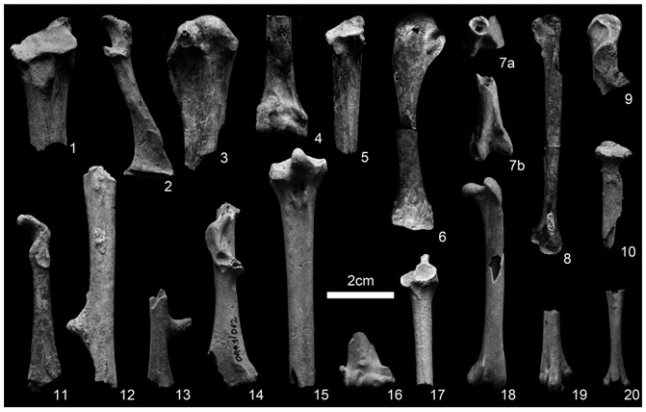


FIGURE 2 Bird bones from studied sites. 1–5: bones from Ban Hua Ud, 6–10: from Ban Non Wat, 11–17: from Keed Kin, and 18–20: from Long Long Rak. 1, 9, 14, and 15: storks, 2–5: cormorants, 6–8, 11–13, and 18–19: pheasants/fowls, 16 and 17: crows, 10: passerine, and 20: taxa undetermined immature bird

of 11 phalanges, four bone fragments (including one immature bone), and an immature tarsometatarsus. The phalanges were also similar in size to those of a modern female red junglefowl specimen. Neither cut nor manufacturing marks were observed on any of the bones.

3.3 | Ban Non Wat

Eighteen of 44 bird remains were identified as pheasant/fowl, stork, and passerines (Passeriformes). Pheasant/fowl was dominant in all of the periods (NISP = 16): all of the four identified bones from the Neolithic, six of seven bones from the Late Neolithic–Early Bronze Age, all of the three bones from the Iron Age, and three of the four identified bones from the Neolithic–Iron Age. One fragment each was observed for stork (Late Neolithic–Early Bronze Age) and passerine (Neolithic–Iron Age). The stork coracoid was similar in size to an Asian openbill specimen (*Anastomus oscitans*) stored at the TNHM. Undetermined fragments included 16 immature bones, which were from the Neolithic–Iron Age. Neither cut nor manufacturing marks were observed on any of the bones.

Four of the 16 fragments identified as pheasant/fowl were tarsometatarsi. There was no medial plantar crest and spur on one of them, whereas it was unknown whether there were crests and spurs on the other. Most of the pheasant/fowl bones were within the size range of male and female red junglefowl specimens, but some of the samples were much larger. Medullary bone was observed at one femur from the Late Neolithic–Early Bronze layer. Six of the bones (three from the Neolithic, one from the Late Neolithic–Early Bronze Age, one from the Iron Age, and the other from the Neolithic–Iron Age) were previously analysed by Storey et al. (2012), and chicken mitochondrial DNA sequences were obtained from two of them. Both of the bones were humeri: one (THABNW003, haplotype: ah2, from the Neolithic layer) was similar in size to a female red junglefowl specimen examined by one of the authors (P.A.), whereas the other (THABNW009, haplotype: ah12, from the Iron Age layer) was similar in size to a male red jungle fowl specimen, owned by P.A. A coracoid sampled by Storey et al. (2012; THABNW006, failed in DNA analysis) was not from pheasant/fowl but from stork (Figure 2, No. 9).

3.4 | Keed Kin

Thirty of 43 bird remains were identified as pheasant/fowl, stork, and crow (Corvidae). Pheasant/fowl was dominant (NISP = 25, MNI = 6), whereas the NISPs (and MNIs) of stork and crow were three (one) and two (one), respectively. Stork tarsometatarsi were similar in size to a specimen of painted stork stored at the TNHM, whereas the crow bones were slightly smaller than a specimen of carrion crow (*Corvus corone*) of M.E. (EP-32). Three phalange and 11 fragmented bones including an immature bone were undetermined. Manufacturing marks were not observed on the bones.

In the bones from pheasant/fowl, hindlimb bones were dominant: tarsometatarsi and tibiotarsi represented 12 and six fragments, respectively. Seven of the 12 tarsometatarsi did not have medial plantar crests, whereas it was unknown whether there were crests on the

others. Spurs were observed in five of the tarsometatarsi, whereas no spurs were observed in four of the tarsometatarsi. Two of the tibiotarsi and one of the tarsometatarsi were immature. Cut marks were observed in the mid-shaft of one tarsometatarsus near a spur. A large-size variation was observed in the pheasant/fowl bones: Some of the bones were smaller than a female red junglefowl specimen whereas others were larger than a male specimen.

4 | DISCUSSION

4.1 | The history of bird exploitation in Thailand

In total, we identified 62 of 117 (52.9%) bird remains from four archaeological sites dating from the Neolithic (4,000–3,000 BP) to the early historic period (fifth–11th Century AD) in Thailand. The analysed bird bones included at least five avian families from four orders: pheasant/fowl (Phasianidae), stork (Ciconiidae), cormorant (Phalacrocoracidae), crow (Corvidae), and passerine (Passeriformes). Pheasants/fowls were dominant in the NISP and were found from three sites, with the exception of Ban Hua Ud, suggesting they were the major bird fauna used in Thailand at least after the Iron Age. Although pheasants/fowls were not found at Ban Hua Ud, a tibiotarsus was found in Layer 4 (Pleistocene–Holocene transition; 10,472–8,345 BP and 12,331–9,215 BP) of Spirit Cave (Mae Hong Son Province [Conrad, Higham, Eda, & Marwick, 2016]). This suggests that the exploitation of pheasants/fowls began in the early Holocene. Cormorants were the second most abundant NISP but were only found at Ban Hua Ud. Storks were the third most abundant NISP but were found at three sites, with the exception of Long Long Rak, where only bones from pheasants/fowls were found. This might relate to environmental factors, as it is the only nonlowland site and is not located near a water source. Cormorant and stork bones were also reported at Khok Phanom Di (Chonburi Province, Neolithic, 4,000–3,500 BP; West, 1991). West (1991) identified bones from great cormorant (*P. carbo*), pygmy cormorant (*Microcarbo pygmaeus*), painted stork (*M. leucocephala*), stork (*Ciconia* sp.), teal (*Anas crecca*), Oriental darter (*Anhinga melanogaster*), pelican (*Pelecanus* sp.), great-billed heron (*Ardea sumatrana*), ibis/spoonbill (Threskiornithidae), sarus crane (*Antigone antigone*), godwit (*Limosa limosa/lapponica*), dusky broadbill (*Corydon sumatranus*), crow (*Corvus* sp.), and junglefowl/domestic fowl (*G. gallus*). Medium to large water birds seem to have been frequently exploited in the lowland area of Thailand since the Neolithic period. There is no evidence for anthropogenic modification of the specimens, whether through butchering or for tool manufacture and secondary use, etc. This is consistent with the fact that there were no bone tools made from bird bones at any of the sites (Shoocongdej, Chintakanon, Pureepatpong, Phunmijumnong, & Saminpanya, 2017). A large number of bone tools made from bovids were found at Keed Kin (Department of Archaeology, 2009). In addition, bone tools made from mammals were also found at other sites, including the late Pleistocene Moh Kiew Cave in southern Thailand (Pookajorn, 1996), suggesting that mammal bones had been exploited as materials for tools since then. Birds seem to have been exploited to meet the demands for food and/or feathers but not as material for bone tools. During the same periods, bone tools made from bird bones (bone spear, needle, and needle case) were common, and many bird bones with manufacturing marks have been found in South China (Eda, Matsui, & Sun, 2016).

Bird bones were also found in occupation layers at Tham Phaa Can (Mae Hong Son Province, 7500 ± 160–5180 ± 110 14C BP) and Tham Boong Hoong (Mae Hong Son Province, 5360 ± 120–930 ± 80 14C BP); however, they were identified as "avian bones" because of the lack of appropriate comparative material (Conrad, 2018). Undetermined large and small "avian bones" were also reported at Ban Na Di (Higham & Kijngam, 1984). It is obvious that the results in this study are regionally and temporally restricted, but they give important quantitative information on bird exploitation during the Neolithic and early historic period in Thailand.

4.2 | The history of chicken exploitation in Thailand

Pheasant/fowl bones were dominant in the NISP in this study and were also found at Spirit Cave (Conrad et al., 2016), suggesting that the exploitation of pheasants/fowls dates back to the early Holocene in this region. Bones from junglefowl/domestic fowl (*G. gallus*) were reported at Khok Phanom Di (West, 1991), whereas bones from domestic fowl were reported at Ban Non Wat (Higham & Kijngam, 2012) and Ban Na Di (Higham & Kijngam, 1984). However, it is important to note that none of the previous studies established morphological bone identification criteria of red junglefowl/domestic fowl from the 24 indigenous Phasianidae species in Thailand, and thus, the identification result is difficult to regard as fully authorized. In this study, we tried to discriminate candidate chicken bones from nonchicken bones by the femora and tarsometatarsi, according to the identification criteria of Eda and Inoué (2011).

We found a femur without pneumatic foramina of the greater trochanter (from Long Long Rak) and 11 tarsometatarsi without medial plantar crests (one from Ban Non Wat, three from Long Long Rak, and seven from Keed Kin). Femora with pneumatic foramina of the greater trochanter or tarsometatarsi with medial plantar crests were absent. Therefore, all of the analysed bones were designated as candidate chicken bones. Fortunately, Storey et al. (2012) succeeded in retrieving mitochondrial DNA sequences of two humeri from Ban Non Wat (one each from the Neolithic and Iron Age) and determined that both of them were from chicken/red junglefowl. It is obvious that mitochondrial DNA haplotypes remained unchanged in the early stages of the domestication process. In our morphological analysis, the humerus from the Neolithic layer was similar in size to that of a female red junglefowl specimen, whereas the other from the Iron Age was similar in size to that of a male red junglefowl specimen. Therefore, there is no evidence to judge whether the exploited birds were from domesticated chickens or wild red junglefowl. Although large Phasianidae birds were found at Ban Non Wat and Keed Kin, the samples could be from indigenous wild pheasants in the region. At this moment, it is difficult to confirm when chickens were domesticated in Thailand, but at least red junglefowls seemed to have been exploited in the late Neolithic, although direct radiocarbon dating of bones is required to eliminate the possibility of their redeposition or contamination from later deposits.

In our previous study of North and central China, both immature Phasianidae bones and candidate chicken bones were rare but were found together at two sites from the Neolithic and Bronze Age, supporting the scenario of captive populations been kept (Eda, Lu, et al., 2016). Immature pheasant/fowl bones were found at Keed Kin, whereas 16 candidate chicken bones and two undetermined immature bones were also found in Ban Non Wat and Long Long Rak, respectively, although it is difficult to identify immature bird bones by morphological analysis. In addition, a pheasant/fowl femur with medullary bone was found at Ban Non Wat. Ancient DNA identification of these immature bones is important, as the presence of significant numbers of immature birds is more likely to be found in captive populations. Further studies combining ancient DNA, morphological analyses, and radiocarbon dating are required to judge whether candidate chicken bones and immature bones are red junglefowl or chicken bones.

It is interesting to note that at Long Long Rak Cave, the bird bones (including candidate chicken bones) were all found inside log coffins along with pig and dog bones as part of grave food. The three lowland sites studied were used as habitation and burial sites, and pheasants/fowls were found at two of those sites with many other species of animals, indicating different human cultural and exploitation patterns in the different environments. Pheasants/fowls exploited in the early Holocene were not a major meat source like pig or cattle; though previous studies have suggested that chickens were domesticated since the Bronze Age at Ban Non Wat and Ban Na Di, the numbers were still quite small in comparison with those of other domesticated animals. Nevertheless, it may have been a major food source in other historical periods, relating to changes in food patterns or preferences. Further comparative zooarchaeological bird bone studies from different periods and ethnozoological research are required to reach further conclusions.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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